GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 04



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Recap of Previous Lecture



Topic Averages

Topics to be Covered



Topic-1 More on Averages

Topic-2 Understanding Percentages

Topic-3 Questionnaire

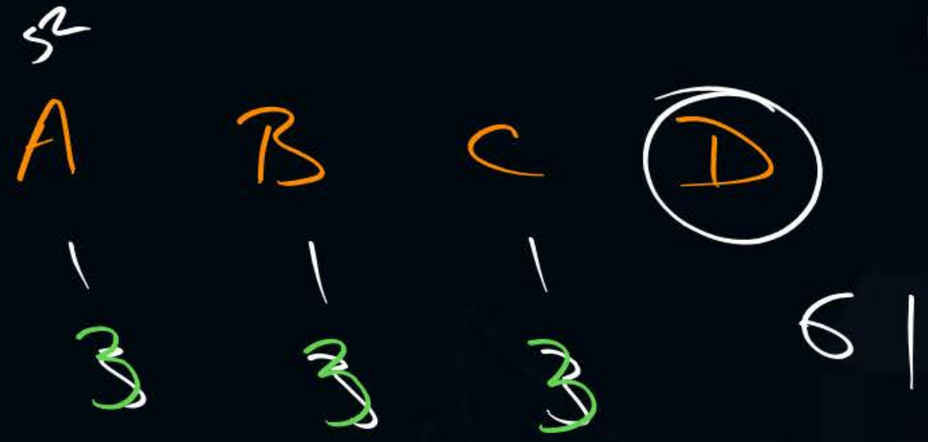
[MCQ]

10×61 =

40×52 = ?

#Q. The average score of a class of 40 students is 52. What will be the average score of the rest of the students if the average score of 10 of the students is 61.

- A 47
- B 49
- C 50
- D 48



Assignment

[MCQ]

#Q. The average weight of a school of 40 teachers is 80 kg. If, however, the weight of the principle be included, the average decreases by 1 kg. What is the weight of the principal?

Assignment

A 49

B 109

C 39

D 29

[MCQ]

#Q. The average age of Abhijeet and Daya is 20 years. Their average age 5 years hence will be

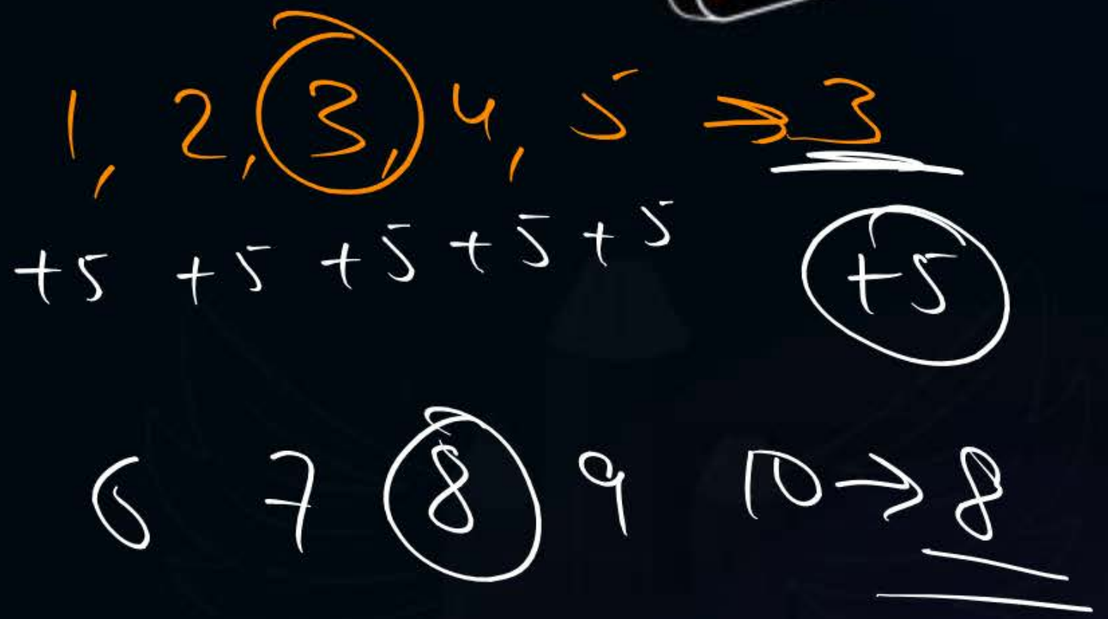
A 20

B 30

C 25

D 22

Assignment



[MCQ]

#Q. The average monthly salary of 20 employes is ₹1500. If the manager's salary is added the average becomes ₹1600. The manager's salary is

A ₹ 3500

B ₹ 3600

C ₹ 3800

D ₹ 3900

[MCQ]

#Q. Three years ago, the average age of a family of 5 members was 17 years. A baby having been born, the average of the family is the same today. What is the age of the baby?

$5 \times 20 = 100$

$6 \times 17 = 102$

2 yrs

A. 6 months

B. 9 monthjs

C. 1 year

D. 2 years

[MCQ]

#Q. 12 yrs ago, the average age of a husband and his wife was 20yrs. The average age is same today, they having two children. What is the present age of the youngest child if children differ in age by 2yrs?

A 8

B 6

C 7

D 9

$H+W=32\times2=64$

$H+W+C_1+C_2=20\times4$

$H+W+C_1+C_2=80$

$C_1+C_2=16$

A 32

B 32

$32-24=8$

7yrs 9yrs

$C_1-C_2=2$

[MCQ]

#Q. The average of 5 consecutive integers starting with x is y. What is the average of 6 consecutive numbers starting with (x+2)?

A $y+3$

B $\frac{2y+9}{2}$

C $y+2$

D $\frac{2y+5}{2}$

$\frac{y+2+y+3}{2}$

$\frac{2y+5}{2}$

$x, x+1, x+2, x+3, x+4 \Rightarrow y$

$x+2=y$

$x+2, x+3, x+4, x \text{----}$

$y, y+1, y+2, y+3, y+4, y+5$

[MCQ]

A

#Q. A cricketer has certain average of runs for his 64 innings. In his 65th innings, he is bowled out for no score on his part. This brings down his average by 2 runs. His new average is?

A 130

B 128

C 70

D 68

130

$$\frac{64A+0}{65} = A-2$$

$$64A+0 = 65A-130$$

$$\Rightarrow A = 130$$

130 − 2 = 128

130 − 2 = 128

PERCENTAGE

For easy 100

100

$\frac{x}{y} \times 100$ → %

10%

53/94

39/60

50% > 33$\frac{1}{3}$%

$\frac{1}{2} > \frac{1}{3}$

$\frac{3}{6} > \frac{2}{6}$

0.5 > 0.3

PERCENTAGE

fraction

1100

50% of 2200

$\frac{50}{100} \times 2200$

$= 1100$

PERCENTAGE

$100\% = 1 = 1$

$5\% = \frac{1}{20} = 0.05$

$10\% = \frac{1}{10} = 0.1$

$15\% = \frac{3}{20} = 0.15$

$20\% = \frac{1}{5} = 0.2$

$25\% = \frac{1}{4} = 0.25$

$30\% = \frac{3}{10} = 0.3$

30%

$33\frac{1}{3}\% = \frac{1}{3} = 0.\overline{3}$

35%

$120\% = \frac{6}{5} = 1.2$

65%

$66\frac{2}{3}\% = \frac{2}{3} = 0.\overline{6}$

70%

PERCENTAGE

Consecutive
Successive

KOOTONS

50% ↓ + 20% ↓ = 60% ↓

0.5 × 0.8 = 0.4

0.60

33.1/3% ↑ 33% ↓

50% ↓ + 50% ↓ = 75% ↓

0.5 × 0.5 = 0.25

0.25

0.75

IBM

10% ↑ + 10% ↑ + 10% ↑ = 33.1% ↑

1.1 × 1.1 × 1.1 = 1.331

25% ↓ = 50% ↑ 50% ↓

1.5 × 0.5 = 0.75

[MCQ]

#Q. When the price of mobile reduced by 20%, the number of mobile sold increased by 40%. The effect on the sale was?

A. 12% increase

B. 12% decrease

C. 32% increase

D. 40% decrease

0.8×1.4

$= 1.12$

12% ↑

of

का

$\frac{1}{n+1}$ of $\frac{2}{n+2}$

$\times$

[MCQ]

#Q. A trader offers three successive discounts of 20%, 10% and 5% to a customer. How much is overall single discount?

Assignment

A 30%

B 31.6%

C 35%

D 68.4%

[MCQ]

#Q. 8% of the people eligible to vote are between 20 and 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25 actually voted. In that election number of person between 20 and 25, who actually voted, was what percentage of those eligible to vote?

Assignment

A 4.2%

B 8%

C 6.4%

D 6.8%



2 mins Summary

Topic Percentages

Successive

'x'

fraction

Decimal

Comparison

THANK - YOU